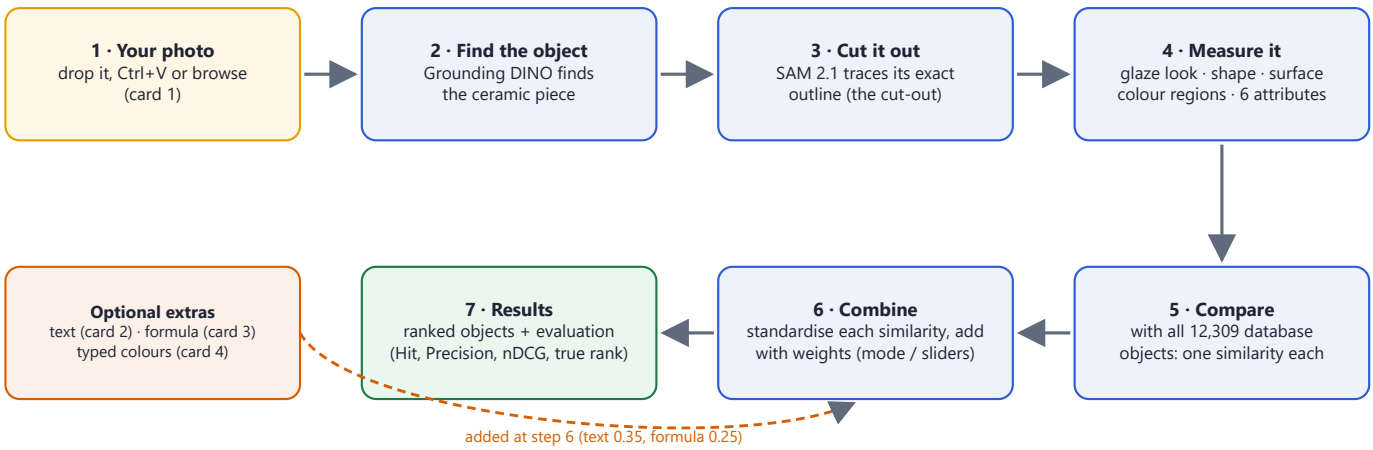
Glaze Search — a visual guide for beginners

What every part of the search page does, how the ranking is made, how each search is evaluated — and why your pasted photo was not found (with the fix). Updated for version 4 (object selection on the photo, HEX picker, priority sliders, evaluation by criterion). Every screenshot is from the real page; every number was measured on 2026-10-20.

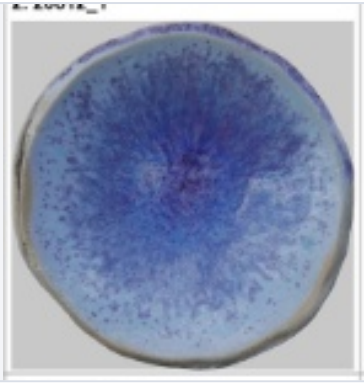
1. The search in one picture
2. Your case: "I pasted the exact image but it was not shown"
3. What is new in version 4
4. The page map
5. Card 1 · Photo: choosing the object
6. Several objects in one photo
7. Card 1 · Pick colour (HEX)
8. Card 1b · Your object (regions, six attributes, (i))
9. Card 2 · Text
10. Card 3 · Formula
11. Card 4 · Colours (and what ΔE means)
12. The search bar: presets and priority sliders
13. How the final score is made
14. Reading the results
15. The evaluation of each search
16. Hit Rate, Precision, Recall and nDCG
17. Session evaluation
18. The (!) and (i) explanations
19. Cheat sheet: what to do for each goal

1 · The search in one picture



The database holds **12,309 objects** (cut out of 7,063 recipe photos + 115 bonus photos). The search measures your object the same way they were measured, gives every database object a similarity for each measurement, then **adds the similarities with weights**. The weights depend on the **mode and priority sliders** and on what else you entered (text, formula, ticked attributes) — section 13 shows exactly how, because that is what went wrong in your search.

2 · Your case: "I pasted the exact image but it was not shown"



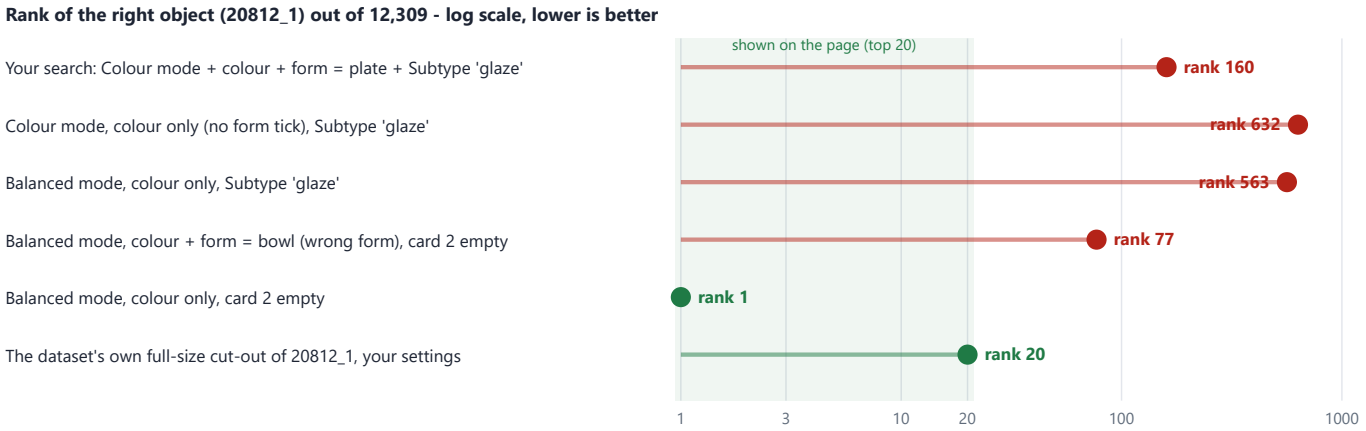
What you pasted: 147 × 153 pixels, a cell cut from one of the contact sheets (grey frame + label "2. 20812_1").



What the page claimed: "same as database photo #666131" — a red-brown tile. Wrong: a false match of the quick photo fingerprint (dHash, 64 bits) on a tiny image. **Fixed** (see below).

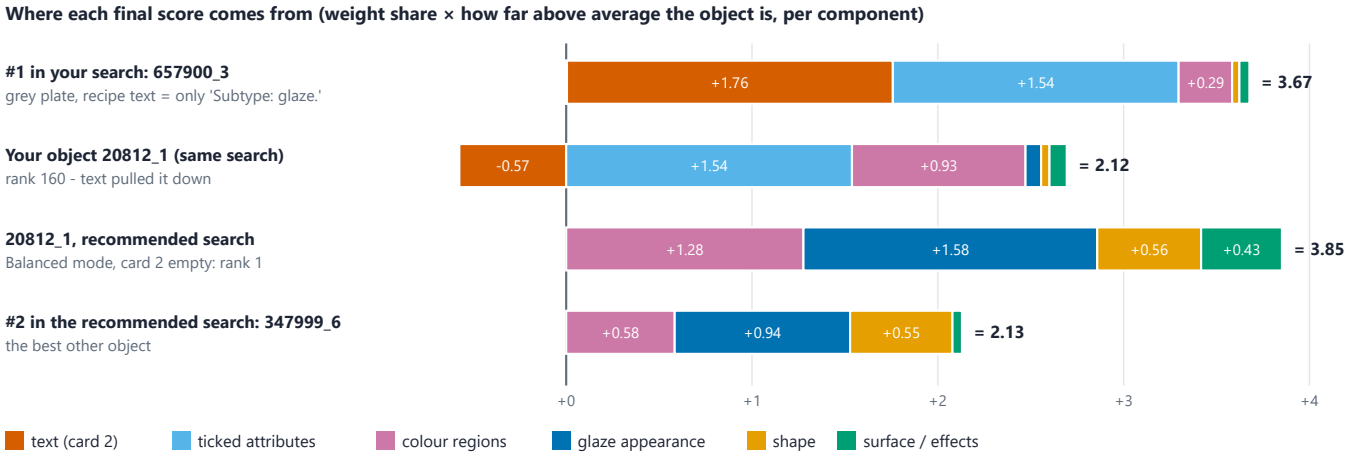


What it really is: object **20812_1** of recipe #20812 "Prickly Potter Peacock" — in the database, searchable.



Your settings were: **Colour** mode, Subtype **"glaze"** typed in card 2, **colour** and **object form = plate** ticked, Top K 20. The right object came **160th**. With card 2 empty and the default Balanced mode it comes **1st**.

Why: the word "glaze" took almost half of the score



1. **Text was 47% of your score.** Text always weighs 0.35, next to 0.40 for the whole photo. Your text "Subtype: glaze." is **word for word the complete text of 502 recipes** (recipes with no cone, surface or description). Those all score text 1.00 and jump to the top — that is why every result said "cone ? · surface ? · transparency ?". Recipe 20812 has a rich description, so its text matched worse (it lost about 2.3 points on text alone).

2. **Colour mode** puts 80% of the photo part on colour regions and almost nothing on the glaze look, where your object was the best match (glaze appearance 0.88 against about 0 for the winners).
3. **Ticking "object form = plate"** is fine here (20812_1 is labelled plate), but the ticked attributes together take 40% of the photo part — with a wrong form (bowl) the object drops to rank 77 even with card 2 empty.
4. **A tiny, framed crop** (147 px with a grey frame and a label) looks less like the original: the glaze-look similarity is 0.88 instead of 1.00, and the cut-out includes part of the frame. With the dataset's full-size cut-out and your settings the object still only reaches rank 20: the text effect stays.

What changed in the app (restart RUN_GLAZE_SEARCH.bat to load it)
· The "database photo" note now needs the pixels to match too (same shape, grey-level correlation ≥ 0.95, colour difference ≤ 14/255); your crop is no longer called a database photo, while real database photos are still recognised even resized or re-saved.
· When your text is word for word the text of 25+ recipes, card 2 and the results show ⚠ **too vague** (see section 9).

To find a known glaze from its photo: use the original photo (not a screenshot of a contact sheet), leave card 2 empty, keep **Balanced**, tick only **colour**, and type the recipe number in **True Glazy recipe ID** — the page then tells you its exact rank.

3 · What is new in version 4 (2026-10-20)

New	Where on the page	Section
Hover an object to see it, **click** to choose it, **Ctrl + click** to add more; the **active** object (yellow) is the one searched	card 1, on the photo	5 – 6
Pick colour (HEX): the colour averaged inside the object under the cursor; copy it or add it as a search colour	card 1 → card 4	7
(i) after every attribute value: what the model says about exactly that value (probability, Grounding DINO score, accuracy)	card 1b	8, 18
Priority sliders shape / glaze / colour, always adding up to 1; the mode buttons are presets of the sliders	search bar	12
Contribution bars: why a result is where it is	every result card	13 – 14
Evaluation of every search by shape, glaze / surface, colour, strict (all) and weighted nDCG	results	15
Metric names: the session's "Recall@K" is now **Hit Rate@K**; Precision@K and a real Recall@K are added	results, session	16 – 17
(!) explanations on the page itself	card 1, search bar, results, session	18

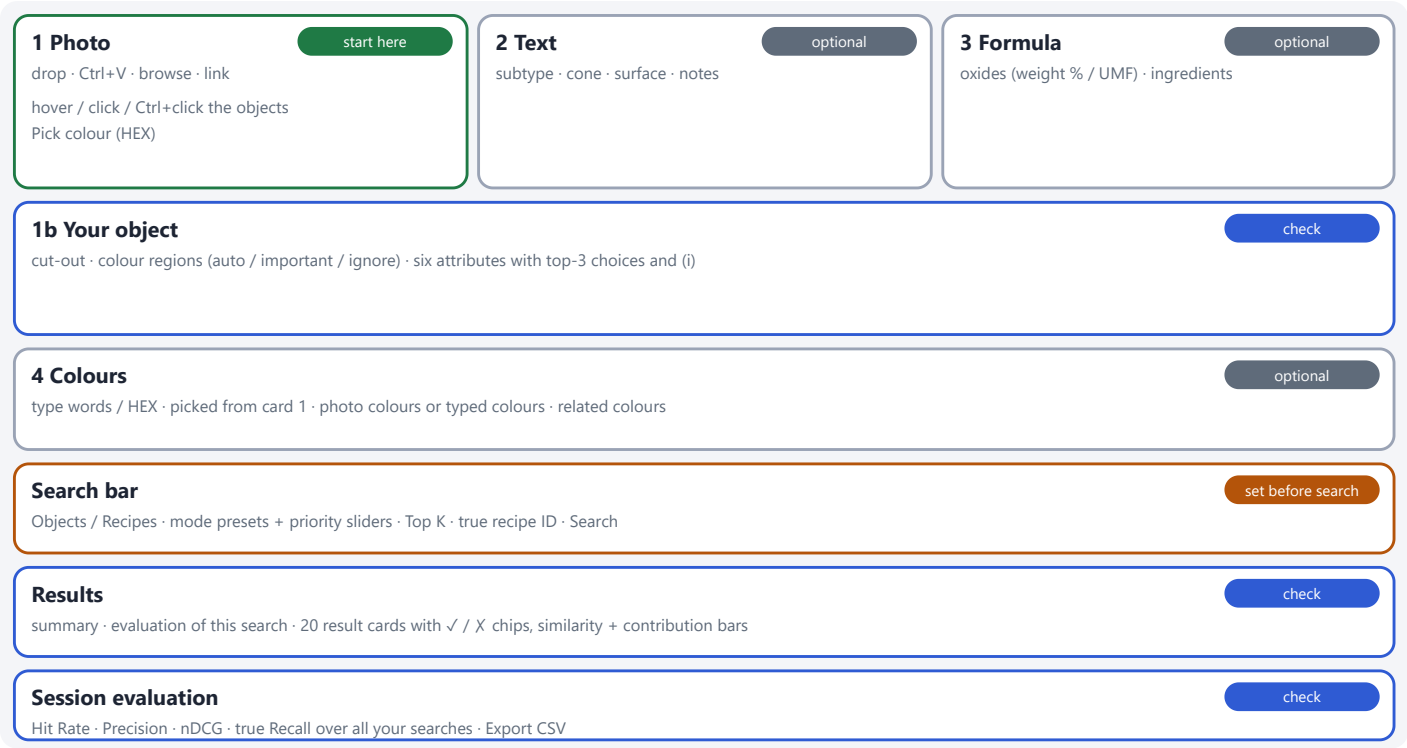
Unchanged on purpose. Each mode button reproduces the earlier validated weights exactly (tested: the same top 20 and the same scores). The evaluation never changes the order. The session log keeps every earlier search, and the CSV keeps its old columns and adds new ones.

Still to come: the definition, reference photos and manual page inside each (i) window. They need your newest reference manuals; until then the window says "Reference pending" instead of inventing a definition.

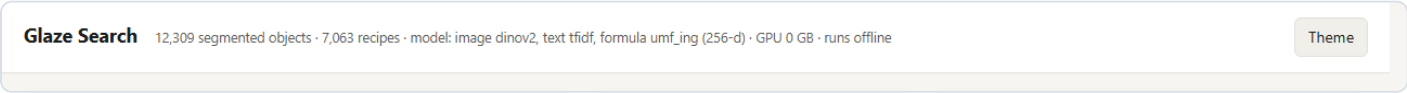
To use version 4, close the Glaze Search window and start `RUN_GLAZE_SEARCH.bat` again.

4 · The page map

The page is a column of cards. You only **need** card 1 and the Search button; everything else refines the search.



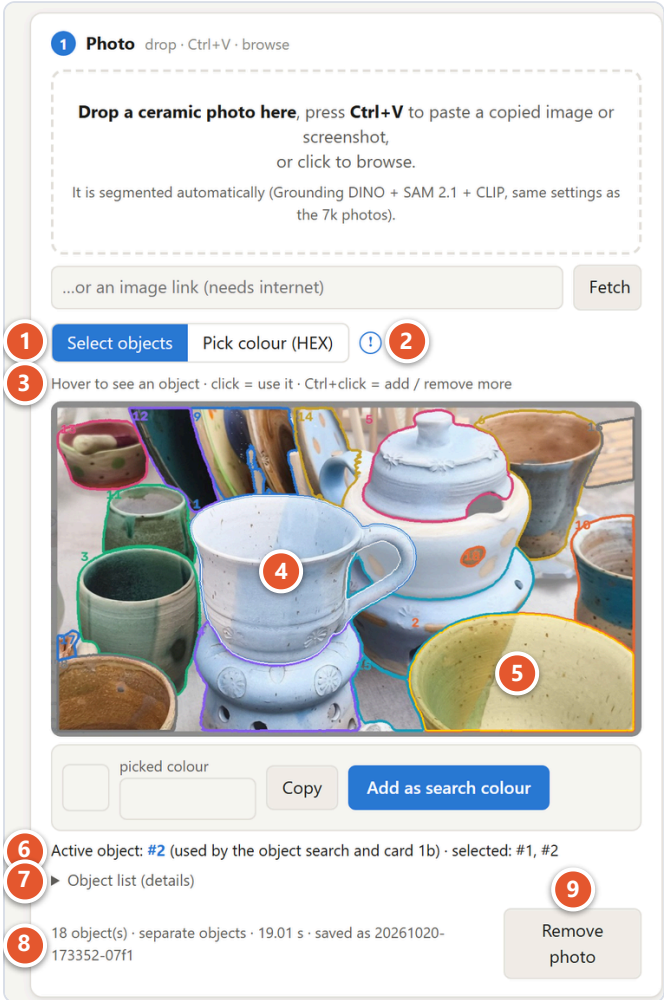
Card	What it is for	When to use it
1 Photo	your photo; the app cuts out the ceramic objects; hover / click to choose one, or pick a colour	always, when you have a photo
1b Your object	check the cut-out; mark colour regions; choose attribute values (top 3, each with (i))	after every new photo — have a look
2 Text	recipe facts you know: subtype, cone, surface, transparency, notes	only with real, specific facts
3 Formula	oxide analysis and ingredients of a recipe	to find recipes with a similar chemistry
4 Colours	colours by words, HEX or picked from the photo, instead of the photo's own colours	"find me a celadon green tile"
Search bar	objects / recipes, mode preset and priority sliders, Top K, the right answer if you know it	before pressing Search
Results	summary, evaluation of this search, 20 ranked objects with ✓ / ✗ chips, similarity and contribution bars	read after each search
Session evaluation	Hit Rate, Precision, nDCG and true Recall over all your searches; Export CSV	when testing how good the search is



The header: database size, the models used, GPU memory, and the Theme button (light / dark).

5 · Card 1 · Photo: choosing the object

A photo can hold several ceramic objects. Each gets an outline and a number; you choose with the mouse **on the photo**.



1. **Select objects** (default) / **Pick colour (HEX)**: what a click on the photo does.
2. **(!)** opens the explanation of the selection on the page.
3. **Hint line**: what hover, click and Ctrl + click do in this mode.
4. **Selected object** #1: white outline (clicked first).
5. **Active object** #2: thick yellow outline and a light yellow fill (Ctrl + clicked last). The search uses this one.
6. **Active object line**: which object is active and which are selected.
7. **Object list (details)**: the same objects as a list, closed by default; a click there also makes an object active.
8. **Status line**: number of objects, the scene type, the time, and the folder where the upload is saved (work/queries/...).
9. **Remove photo**: start again.

Hover: see what an object is before you click



1. **Label**: object number, how sure CLIP is that it is ceramic, its share of the photo and its state (active / selected / click to use).
2. **The object under the cursor** lights up in yellow.
3. **All other objects** are dimmed while you hover.

6 · Several objects in one photo

You do	What happens
Hover an object	it lights up, the rest dims, and a label shows its number, ceramic score and share of the photo
Click an object	it becomes the **active object** (thick yellow outline) and the only selected one
Ctrl + click an object	adds it to the selection and makes it active — or removes it if it was selected; the other selected objects keep a white outline
Object list (details)	the same choices as a list; the framed entry is the active object

What is used where

Part of the page	Uses
Objects search (the default)	only the **active** object: its look, shape, colour regions and attributes are compared with the 12,309 database objects
Card 1b (your object)	the active object — its colour regions and its six attributes
Evaluation of the search (section 15)	the values of the active object: its #1 candidate, or the value you chose in card 1b
Recipes (previous model) search	**all selected** objects together (their image features averaged by area)
Pick colour (HEX)	the object under the cursor, whether it is selected or not

> **Measured example (the shelf photo above, 18 objects):** click #1 (the cup with a handle), then Ctrl + click #2 (the bowl). The page says "*Active object: #2 (used by the object search and card 1b) · selected: #1, #2*". The Objects search, card 1b and the evaluation now use #2. To search for the cup, click it (without Ctrl) and press Search again.

What happens when you add a photo

a. **Grounding DINO** reads the photo with the words "ceramic object, pottery piece, test tile …" and draws boxes.

b. **SAM 2.1** turns each box into an exact outline (a mask); the cut-out is the object without the background.

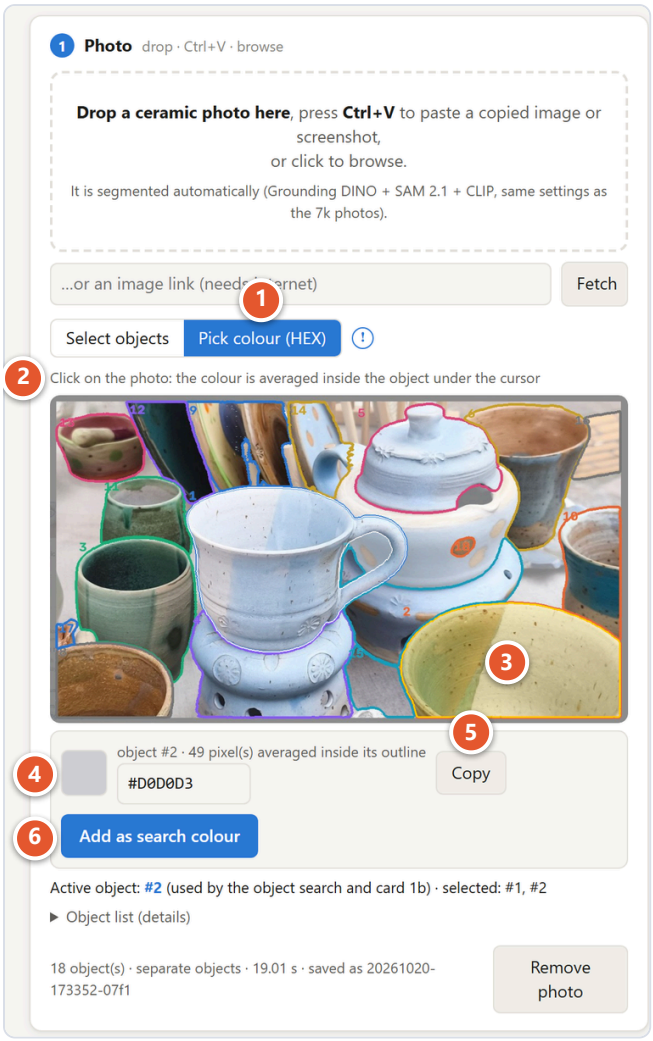
c. **CLIP** checks each one is ceramic (score 0–1); objects ≥ 0.35 are selected and the first is active (otherwise the first object).

d. A quick fingerprint, confirmed pixel by pixel, checks whether the photo is already in the database; if so a note says which recipe, and the search is left out of the session evaluation (it would find itself).

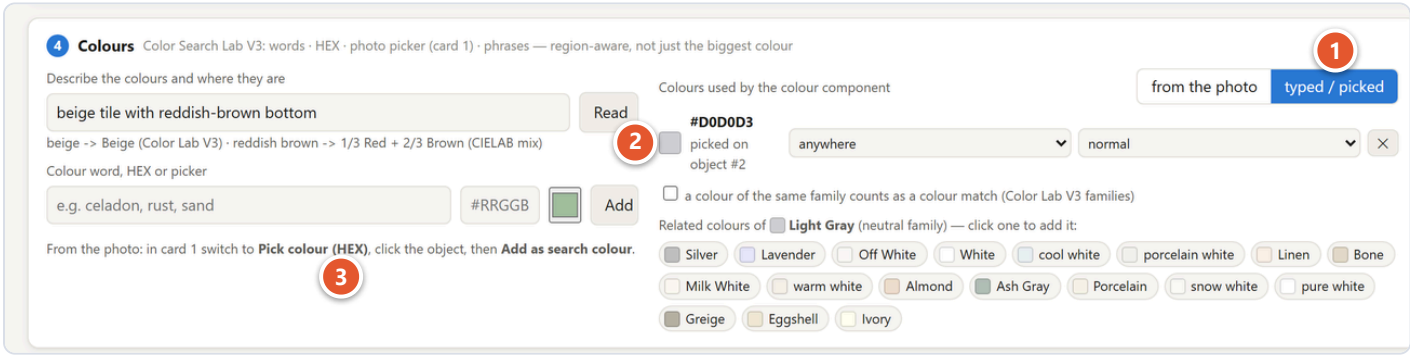
Tips for a good photo

· The object large and sharp, at least ~500 pixels wide; tiny screenshots lose the glaze detail.
· No frames, labels or other objects touching it — they can end up in the cut-out.
· Daylight-like light; strong coloured light changes the colours.

7 · Card 1 · Pick colour (HEX)



1. **Pick colour (HEX)**: a click now picks a colour; the selection stays as it was.
2. Hint line for this mode.
3. **Where you click**: the colour is averaged over a 7 × 7 pixel square, using only the pixels inside the object under the cursor (the background is left out).
4. **The picked colour**: swatch, where it came from (here "object #2 · 49 pixel(s) averaged inside its outline") and the HEX value (#D0D0D3), which you can edit.
5. **Copy** the HEX to the clipboard.
6. **Add as search colour**: puts it in card 4 (picture below).

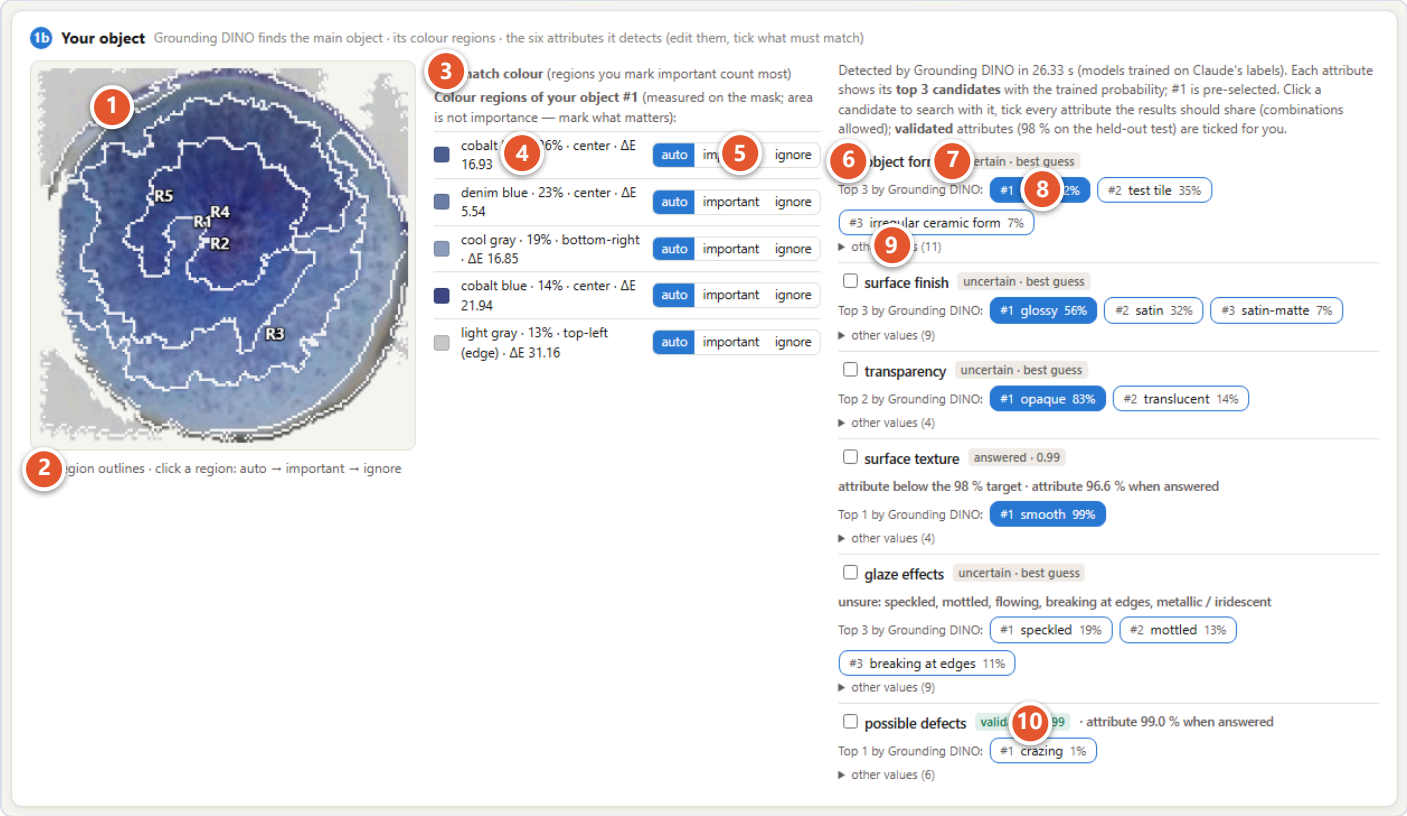


1. Card 4 switches to **typed / picked**: the colour score now uses this colour instead of the photo's colour regions.
2. **The picked colour**, with position and importance like a typed colour; ✕ removes it.
3. The hint in card 4 points to the picker in card 1.

Clicking outside every object still gives a colour, with a warning — the background is rarely what you want. When you search with the photo's own colours again, switch card 4 back to **from the photo**.

8 · Card 1b · Your object

After the photo is cut out, this card shows **what the app measured** and lets you correct it before searching.



1. **The cut-out with its colour regions** R1, R2 … (areas of similar colour). Click a region on the picture to cycle auto → important → ignore.
2. **Region outlines** on / off.
3. **match colour**: tick it when the results should share your colours (✓ colour in the results).
4. **One row per region**: colour name · share of the object · position · ΔE (how different the region is from the rest of the object).
5. **auto / important / ignore**: important regions count fully (1.0), normal half (0.5), ignored not at all; auto lets the app weigh them by size and distinctness. Ignore labels, holes, shadows, bare clay.
6. **Tick box**: use this attribute in the search (results must share it). Combine as many as you like.
7. **Badge**: validated / answered / uncertain - how sure the model is (table below).
8. **Top-3 candidates** from Grounding DINO with the trained probability; #1 is selected. Click the one that is right.
9. **other values**: the full list, when none of the three is right.
10. **validated · 0.99**: confident enough to be ticked automatically.

The six attributes

Attribute	Values (examples)
object form	test tile, bowl, plate, cup, vase, jar, sculpture …
surface finish	glossy, semi-glossy, satin, satin-matte, matte …
transparency	transparent, translucent, semi-opaque, opaque

What the badges mean

Badge	Meaning	Ticked for you?
validated	the attribute reached 98 % correct on held-out test objects **and** this answer is above its threshold	yes
answered	above the threshold, but the attribute as a whole is below	no

surface texture	smooth, rough, granular, dimpled
glaze effects	speckled, mottled, flowing, breaking at edges, crystalline … (several possible)
possible defects	pinholes, blistering, crawling, crazing … (several possible)

	98 %	
uncertain · best guess	below the threshold: look at the top 3 and decide	no
set by you	you clicked a value	yes

> Surface finish, transparency and texture are hard to judge from one photo: the right value is often #2 or #3 — that is why the top 3 are shown. Object form is the most reliable (right first 84 % of the time on the test objects).

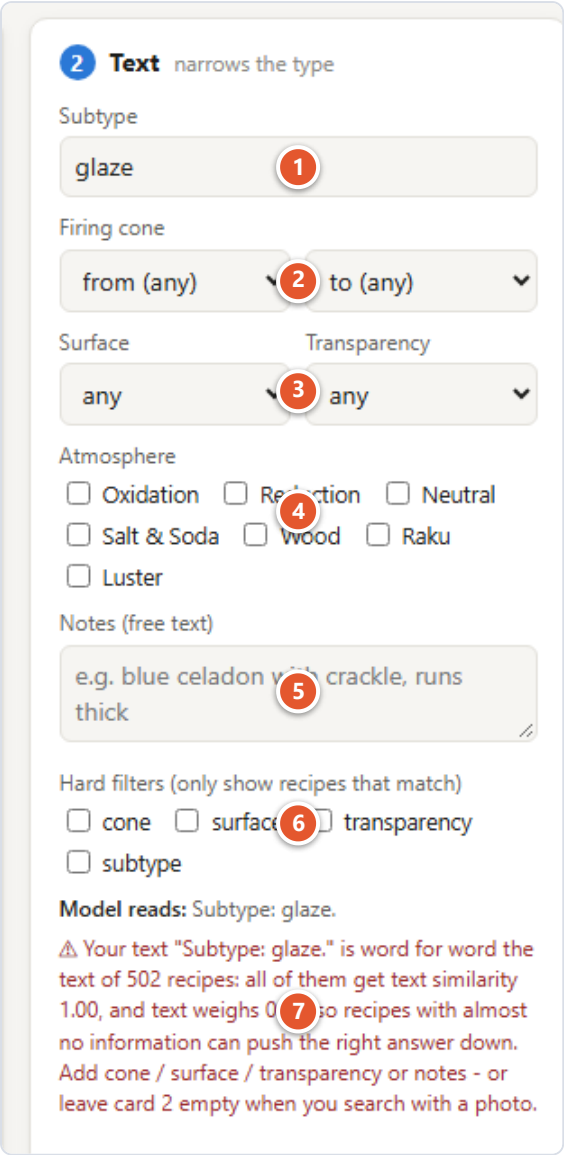
(i): what the model says about one value



1. **Tick box**: use this attribute in the ranking.
2. **Top-3 candidate** with the model probability; the blue one is the chosen value.
3. **(i)**: explains exactly this value for your object (section 18).

> The chosen value (the #1 candidate until you click another) is also what the evaluation checks (section 15), even when its box is not ticked.

9 · Card 2 · Text



1. **Subtype**: the Glazy type, e.g. "glaze - iron - celadon" (suggestions appear as you type).
2. **Firing cone** from / to, e.g. 6 to 6.
3. **Surface** and **Transparency** as written in the recipe (glossy, matte ...; opaque, translucent ...).
4. **Atmosphere**: oxidation, reduction, wood ...
5. **Notes**: free text, e.g. "blue celadon with crackle, runs thick".
6. **Hard filters**: keep only recipes that match exactly - they work in **Recipes (previous model)** search only, not in Objects search.
7. **Model reads**: the sentence the text model actually compares, and the new ⚠ **too vague** warning.

How text is used

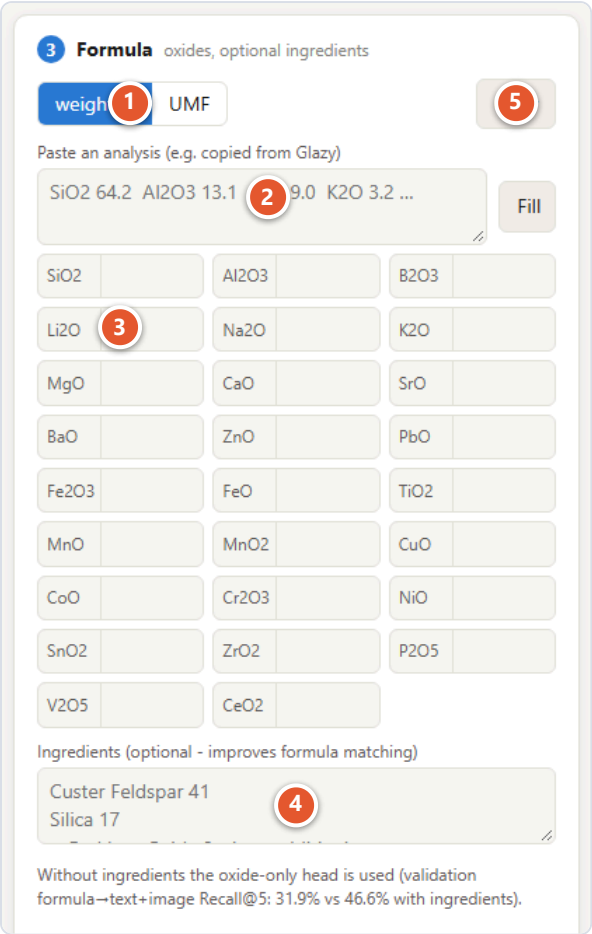
Everything you enter is turned into one short sentence ("Subtype: glaze. Surface: glossy. Firing cone: 6 ..." + notes). The model compares it with the same kind of sentence written from each recipe's Glazy data, and the similarity is added with weight **0.35**.

Your text	A recipe's text	Text similarity
Subtype: glaze.	Subtype: glaze. (502 recipes, nothing else known)	**1.00** — a perfect match that says nothing
Subtype: glaze.	Subtype: glaze. Transparency: opaque. Firing cone: 6. Atmosphere: oxidation. "An opaque white glaze, with red, purple and blue pooling..." (20812)	**-0.32** — the right recipe scores low

Rule of thumb: with a photo, leave card 2 **empty** unless you really know the cone, surface or transparency of the glaze you are looking for. A single generic word ("glaze") makes the recipes with the *least* information win.

Use card 2 alone (no photo) when you only know facts: e.g. Subtype "glaze - iron - celadon", cone 10, reduction, notes "crackle" → recipes described that way.

10 · Card 3 · Formula



1. **weight % / UMF**: how your numbers are written. Weight % = percent of each oxide (as on Glazy's analysis); UMF = the unity molecular formula (fluxes add up to 1).
2. **Paste an analysis** copied from Glazy, then **Fill** puts each number in its box.
3. **Oxide boxes**: SiO2 (silica, glass former), Al2O3 (alumina), fluxes (CaO, MgO, K2O, Na2O ...), colourants (Fe2O3, CoO, CuO ...).
4. **Ingredients** (optional), one per line, e.g. "Custer Feldspar 41". They make formula matching clearly better.
5. **Clear** all formula boxes.

What the formula does

The app converts your oxides into the unity molecular formula (UMF) — the chemist's way of comparing glazes — and finds recipes whose chemistry is close. It is added with weight **0.25**.

> Needs at least **3 oxides including a flux** (Na2O, K2O, CaO, MgO ...); otherwise it is ignored and a note says so.

> With ingredients the formula search is much better: on validation, formula → recipe (text + image) Recall@5 was 46.6 % with ingredients and 31.9 % with oxides only.

Typical use: you have a recipe and want glazes with a similar chemistry and their photos — paste its analysis, leave the photo empty, search.

11 · Card 4 · Colours



1. **Describe the colours and where they are**, e.g. "beige tile with reddish-brown bottom", then **Read**: the app turns it into colours + positions.
2. **Colour word, HEX or picker**: "celadon", "#7FA88A" or the colour box, then **Add**.
3. **From the photo**: use **Pick colour (HEX)** in card 1 (section 7).
4. **from the photo / typed / picked**: which colours the colour score uses. Typed colours **replace** the photo's colours.
5. **Position** (anywhere, top, bottom, centre ...) and **importance** (normal / important) of each colour.
6. ✕ removes a colour.
7. **Same family counts as a match**: e.g. any brown counts for "rust" (wider colour rule).
8. **Related colours** (Color Lab V3): click one to add it.

The phrase above was read as #F5F5DC (beige tile, anywhere) and #C32521 (reddish brown bottom, bottom).

ΔE2000 = how different two colours look (0 = identical)

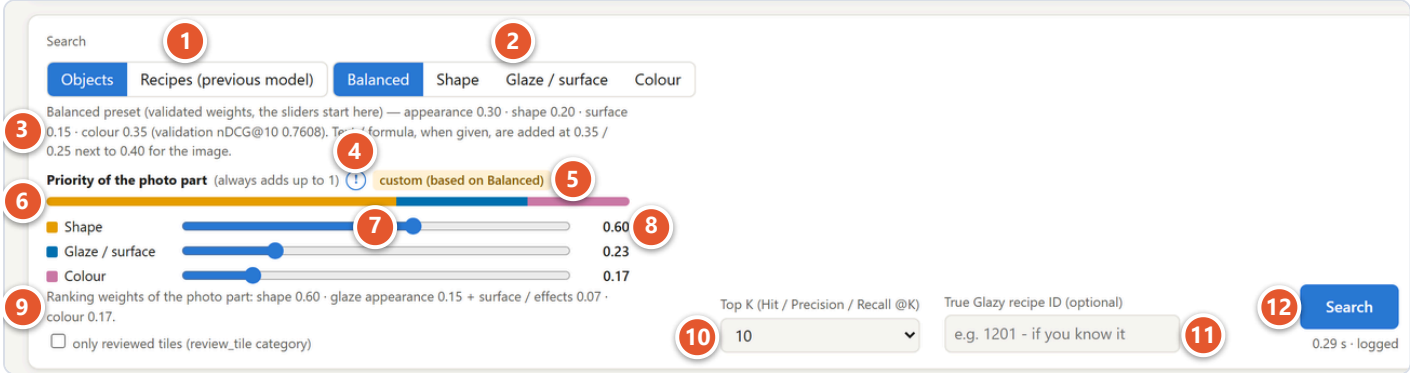


ΔE 1	ΔE 3	ΔE 6	ΔE 10	ΔE 20	ΔE 35
same to the eye	barely visible	noticeable side by side	clearly different shade	the app's colour-match limit	a different colour

In the results, a colour counts as matching (✓ colour) when your most important colour is within ΔE 20 of one of the object's colour regions.

> The colour score compares regions, not just the biggest colour: a tile with a blue top and brown bottom matches "blue top, brown bottom" better than an all-blue tile. Mark what matters as **important** (here or in card 1b).

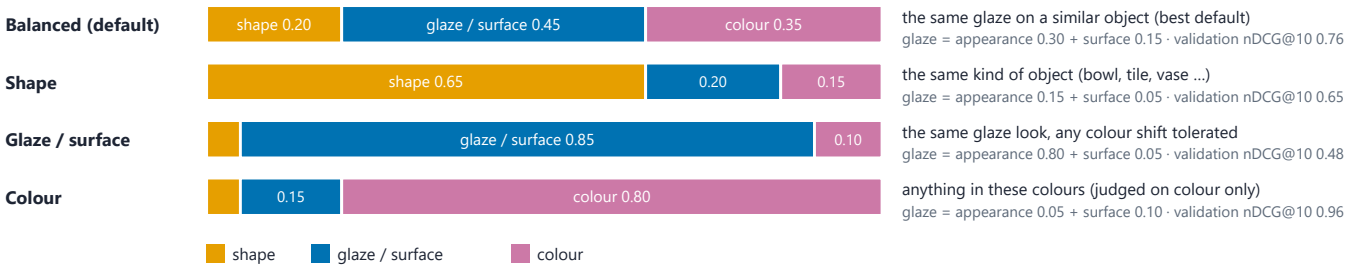
12 · The search bar: presets and priority sliders



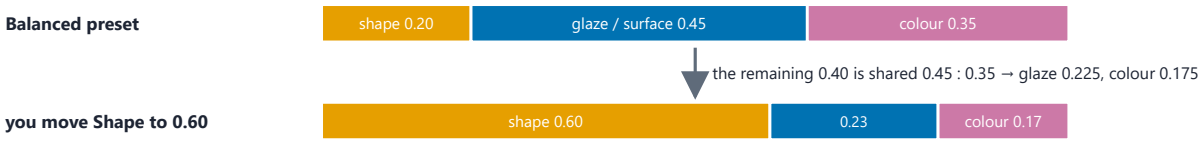
1. **Objects** (default): every cut-out object is a result (12,309). **Recipes (previous model)**: one result per recipe, with image / text / formula weights and the hard filters of card 2.
2. **Mode buttons = presets**: each sets the sliders to its validated weights.
3. The preset's own weights and its validation quality (nDCG@10).
4. **(!)**: the explanation of the priority, with the presets table.
5. **custom**: the sliders differ from the preset (the results then say "custom priority").
6. **Priority bar**: the three shares at a glance.
7. **Sliders** shape, glaze / surface, colour. Moving one rescales the other two, so the total stays 1.
8. The current value of each slider.
9. The ranking weights that follow (glaze is split into appearance + surface / effects).
10. **Top K** for the evaluation: 3, 5, 10, 15 or 20 (the page always lists 20 results).
11. **True Glazy recipe ID**: the right answer, if you know it — gives its rank and the real Recall@K.
12. **Search** (or `Enter`); below it the time and "logged".

The **only reviewed tiles** box limits the search to the objects you put in the review_tile category in the review app.

The mode buttons are presets of the three sliders (each row adds up to 1)



Moving one slider: the other two share the rest in their old proportion



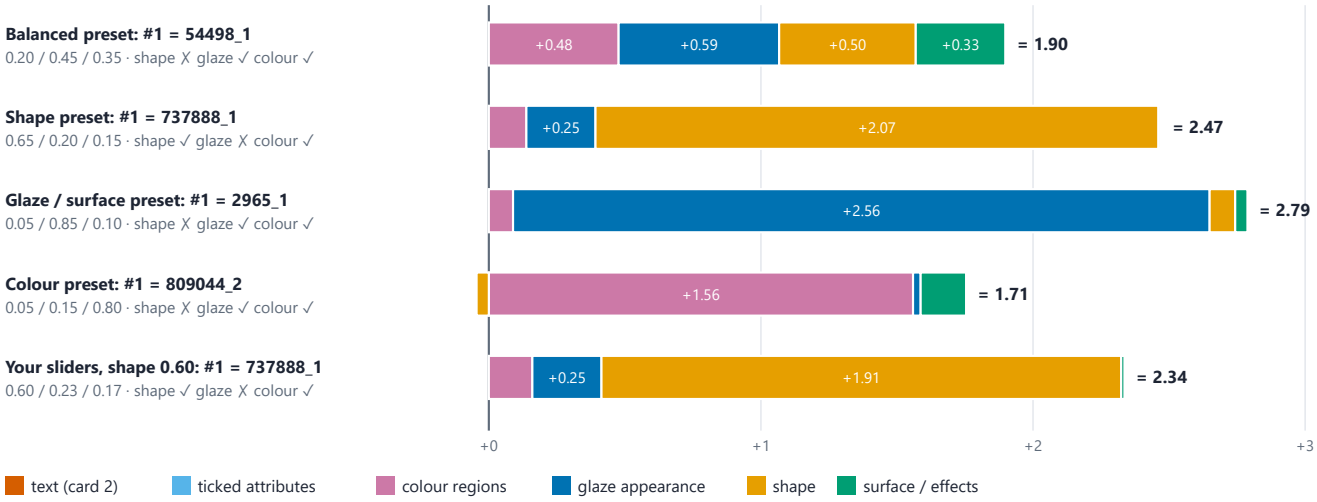
> The presets were tuned on validation searches; your own slider settings were not. Each preset's nDCG@10 uses its own relevance rule, so the numbers are not comparable between presets. The colour slider only counts when **match colour** is ticked in card 1b; otherwise shape and glaze share the photo part.

13 · How the final score is made

1. For each component (glaze look, shape, surface / effects, colour regions, ticked attributes, text, formula) every database object gets a **similarity** to your query — the numbers on the similarity bars of each result card. They are **not probabilities**.
2. Each similarity is **standardised** over all 12,309 objects: how many "standard steps" above the average this object is (a z-score). This makes components with different scales comparable.
3. The standardised values are **multiplied by their weight** and added. The photo part has 0.40 in total, shared by the **sliders**: shape gets the shape slider, glaze appearance + surface / effects get the glaze slider (split like the mode, Balanced 2 : 1), colour regions the colour slider. Ticked attributes take 40 % of the photo part; text adds 0.35 and formula 0.25. All weights are divided by their total.
4. **Contribution** of a component = its weight share × its standardised value. The contributions of a result add up to its final score; the result card shows them as green (pushes up) and red (pushes down) bars.

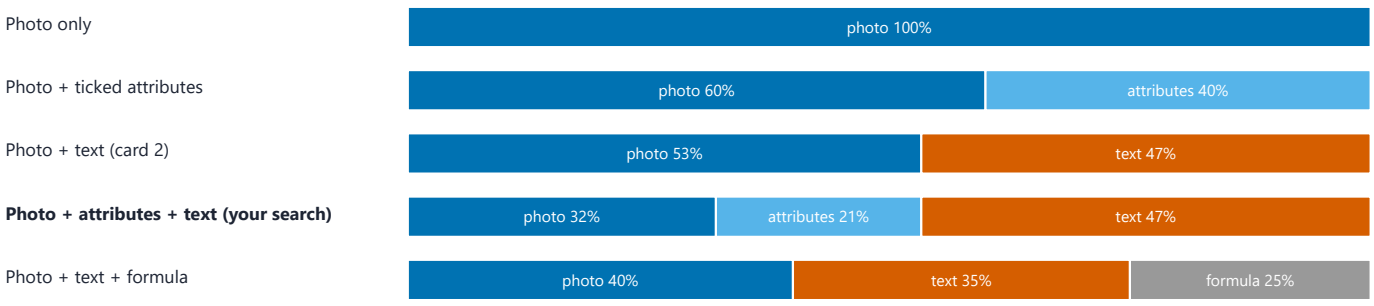
Measured: sliders shape 0.60 · glaze 0.225 · colour 0.175 gave the photo weights shape 0.24 · glaze appearance 0.06 · surface / effects 0.03 · colour regions 0.07 (they add up to 0.40, the photo part).

Same object (#2, a bowl): the #1 result of each setting and where its score comes from



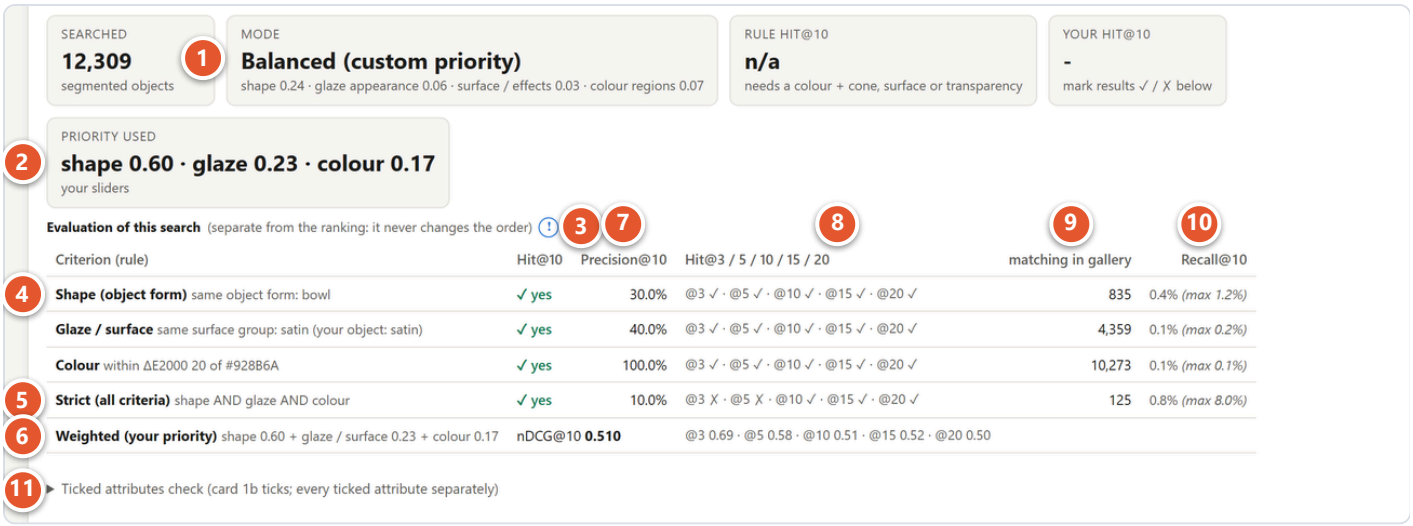
Moving the priority changes which component wins: with the Shape preset or the shape slider at 0.60, shape gives most of the #1 score; with Glaze / surface it is the glaze appearance, with Colour the colour regions.

Share of the final score (each part divided by the total weight)



As soon as you type anything in card 2, text is **about half** of the score with a photo (47 % in the case of section 2). That is right when the text is specific, and harmful when it is a single generic word.

14 · Reading the results

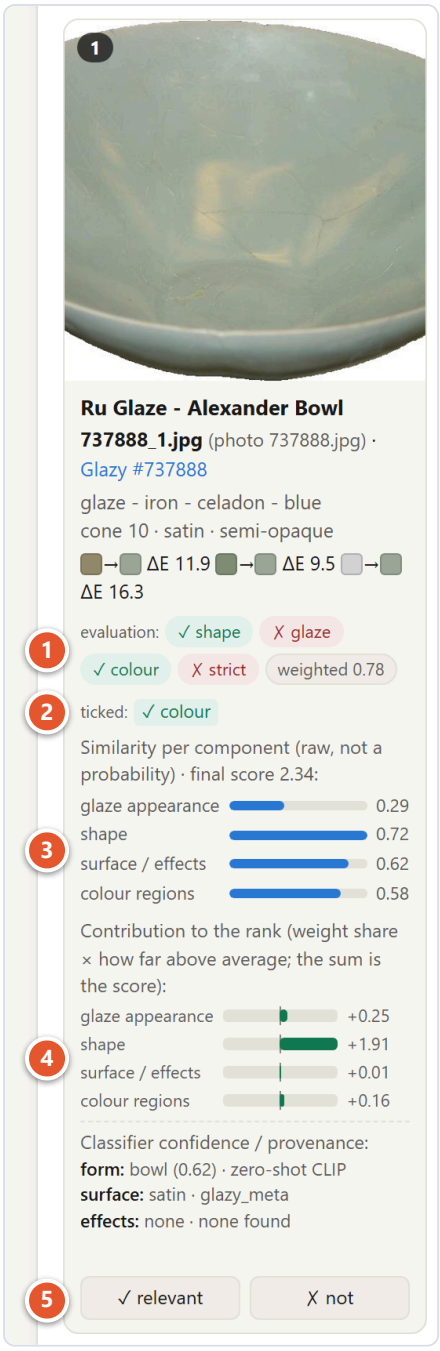


The search with the sliders at shape 0.60 (shelf photo, active object #2, a bowl).

1. **Mode** and the weights used in this search (each divided by their total, so only the ratios matter).
2. **Priority used**: your sliders, or the preset.
3. **(!)**: how the evaluation works.
4. **One row per criterion** — shape, glaze / surface, colour — with its rule (here: same object form: bowl). **Hit@K**: at least one relevant result in the top K?
5. **Strict**: shape AND glaze AND colour.
6. **Weighted**: graded by your priority (nDCG).
7. **Precision@K**: the share of the top K that is relevant.
8. **Hit@3 / 5 / 10 / 15 / 20**: the hit for every K at once.
9. **Matching in gallery**: how many of the 12,309 objects meet the rule (+ how many have no label: ?).
10. **Recall@K** and its maximum (K ÷ matching): tiny when thousands match — read Precision instead.
11. **Ticked attributes check** (folded): every ticked attribute on its own, as before.

The other summary cards are unchanged: **Searched** (objects compared), **Rule Hit@K** (only with a colour and a cone / surface / transparency in card 2), **Your Hit@K** (from your ✓ / ✗ marks) and, with a True Glazy recipe ID, its rank (section 15).

Anatomy of one result card



1. **evaluation:** ✓ / ✗ / ? per criterion, strict, and the weighted relevance (0 – 1). It never changes the order.
2. **ticked:** ✓ / ✗ for each attribute you ticked in card 1b, with the label used and its source.
3. **Similarity per component**: the raw similarity (−1 … 1) — not a probability.
4. **Contribution to the rank**: weight share × how far above average; the sum is the final score.
5. **✓ relevant / ✗ not**: your own judgement (saved to the log).

Above the chips: the **rank**, the recipe name, the object file and its photo, the **Glazy link**, the recipe data (subtype · cone · surface · transparency, "?" = unknown on Glazy) and your colours → the object's closest colours with ΔE. At the bottom: the older labels of the object and where they come from. Click the image to open the whole photo.

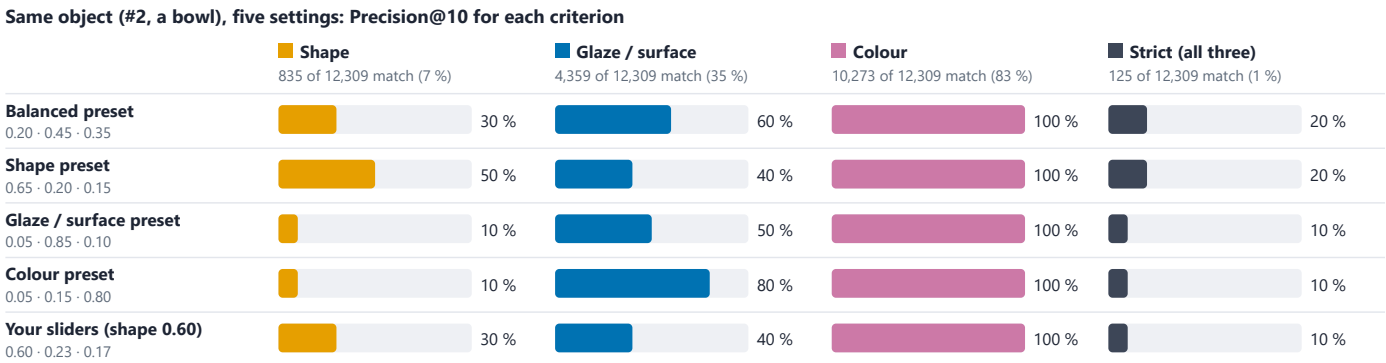
Which label of a database object is used ("source" in the result card) - the first one that exists wins



15 · The evaluation of each search

Every search is checked against the values of **your active object** (its #1 candidate, or what you chose in card 1b). The labels of the database objects come from the best available source: your review > Claude's check > Grounding DINO > the earlier model. The evaluation **never changes the order** of the results.

Criterion	A result is relevant when	Example
Shape	it has the same object form	your bowl → bowls
Glaze / surface	its surface finish is in the same group (glossy / satin / matte); if you picked glaze effects, it also shares one of them	satin-matte counts as satin
Colour	your most important colour is within ΔE2000 20 of one of its colour regions (or in the same colour family, if ticked in card 4)	#928B6A
Strict	all criteria above match at once	bowl + satin + colour
Weighted	your priority × the matches, from 0 to 1	0.20 × 0 + 0.45 × 1 + 0.35 × 1 = **0.80** for the Balanced #1 (54498_1)
?	the object has no label for that criterion: never counted as a match, and counted apart	"+ n ?" in the table

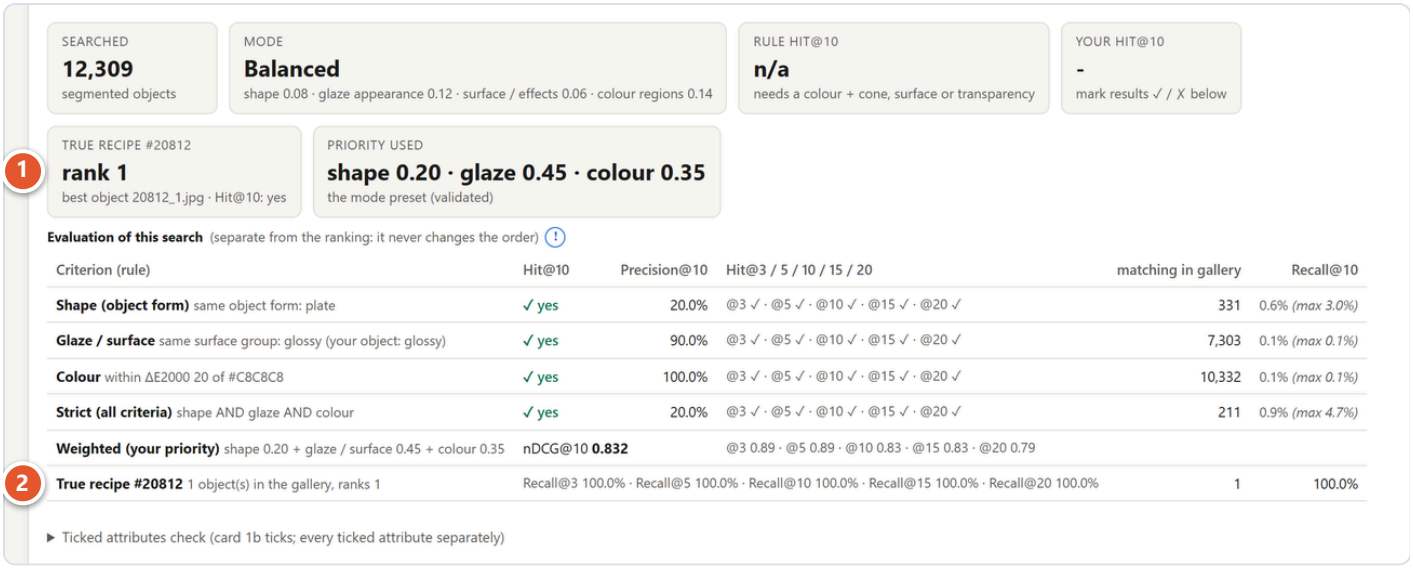


Read "matching in gallery" first. 10,273 of the 12,309 objects (83 %) are within ΔE 20 of #928B6A, so colour is 100 % in every setting: for greyish, neutral colours the colour rule is lenient. Shape (835 bowls) and strict (125) say much more.

One object is a small test. With 10 results, one result more or less is 10 %. Compare settings over many searches in the session table (section 17). Weighted nDCG uses *your* priority as the relevance, so two different priorities answer two different questions — compare them by Precision per criterion, or by the true rank.

The true recipe: the strictest test

When you know the right recipe, type its number in **True Glazy recipe ID**. The evaluation then adds the only real Recall@K: the share of that recipe's objects that reached the top K.



SEARCHED
12,309
segmented objects

MODE
Balanced
shape 0.08 · glaze appearance 0.12 · surface / effects 0.06 · colour regions 0.14

RULE HIT@10
n/a
needs a colour + cone, surface or transparency

YOUR HIT@10
-
mark results ✓ / ✗ below

1

TRUE RECIPE #20812
rank 1
best object 20812_1.jpg · Hit@10: yes

PRIORITY USED
shape 0.20 · glaze 0.45 · colour 0.35
the mode preset (validated)

Evaluation of this search (separate from the ranking: it never changes the order)

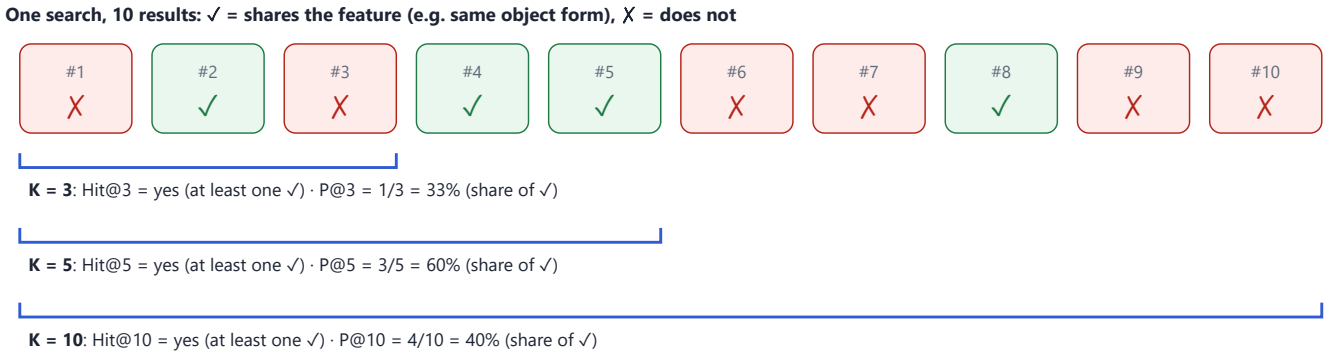
Criterion (rule)	Hit@10	Precision@10	Hit@3 / 5 / 10 / 15 / 20	matching in gallery	Recall@10
Shape (object form) same object form: plate	✓ yes	20.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	331	0.6% *(max 3.0%)*
Glaze / surface same surface group: glossy (your object: glossy)	✓ yes	90.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	7,303	0.1% *(max 0.1%)*
Colour within ΔE2000 20 of #C8C8C8	✓ yes	100.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	10,332	0.1% *(max 0.1%)*
Strict (all criteria) shape AND glaze AND colour	✓ yes	20.0%	@3 ✓ · @5 ✓ · @10 ✓ · @15 ✓ · @20 ✓	211	0.9% *(max 4.7%)*
Weighted (your priority) shape 0.20 + glaze / surface 0.45 + colour 0.35	nDCG@10 **0.832**		@3 0.89 · @5 0.89 · @10 0.83 · @15 0.83 · @20 0.79		
2 **True recipe #20812** 1 object(s) in the gallery, ranks 1	Recall@3 100.0% · Recall@5 100.0% · Recall@10 100.0% · Recall@15 100.0% · Recall@20 100.0%			1	100.0%

▶ Ticked attributes check (card 1b ticks; every ticked attribute separately)

The case of section 2, searched the recommended way (Balanced, card 2 empty, colour ticked, True Glazy recipe ID 20812).

1. **True recipe card**: the rank of the best object of recipe #20812.
2. **True recipe row**: how many objects that recipe has in the gallery, their ranks, and the real Recall@K.

16 · Hit Rate, Precision, Recall and nDCG



Hit Rate@K in the session table = the share of your searches whose Hit@K was "yes" (called Recall@K before 2026-10-20).

Metric	Question it answers	Example (top 10 has 4 relevant results)
Hit@K (one search)	is there at least one relevant result in the top K?	yes
Hit Rate@K (session)	in what share of my searches was Hit@K "yes"?	—
Precision@K	what share of the top K is relevant?	4 / 10 = 40 %
Recall@K	what share of *all* relevant objects reached the top K?	with 835 bowls in the gallery: at most 10 / 835 = 1.2 % — not informative
True recipe Recall@K	what share of the right recipe's objects (usually 1 – 3) reached the top K?	recipe 20812 has 1 object, at rank 1 → 100 %
nDCG@K	are the most relevant results (weighted by your priority) near the top? 1 = the best possible order	—

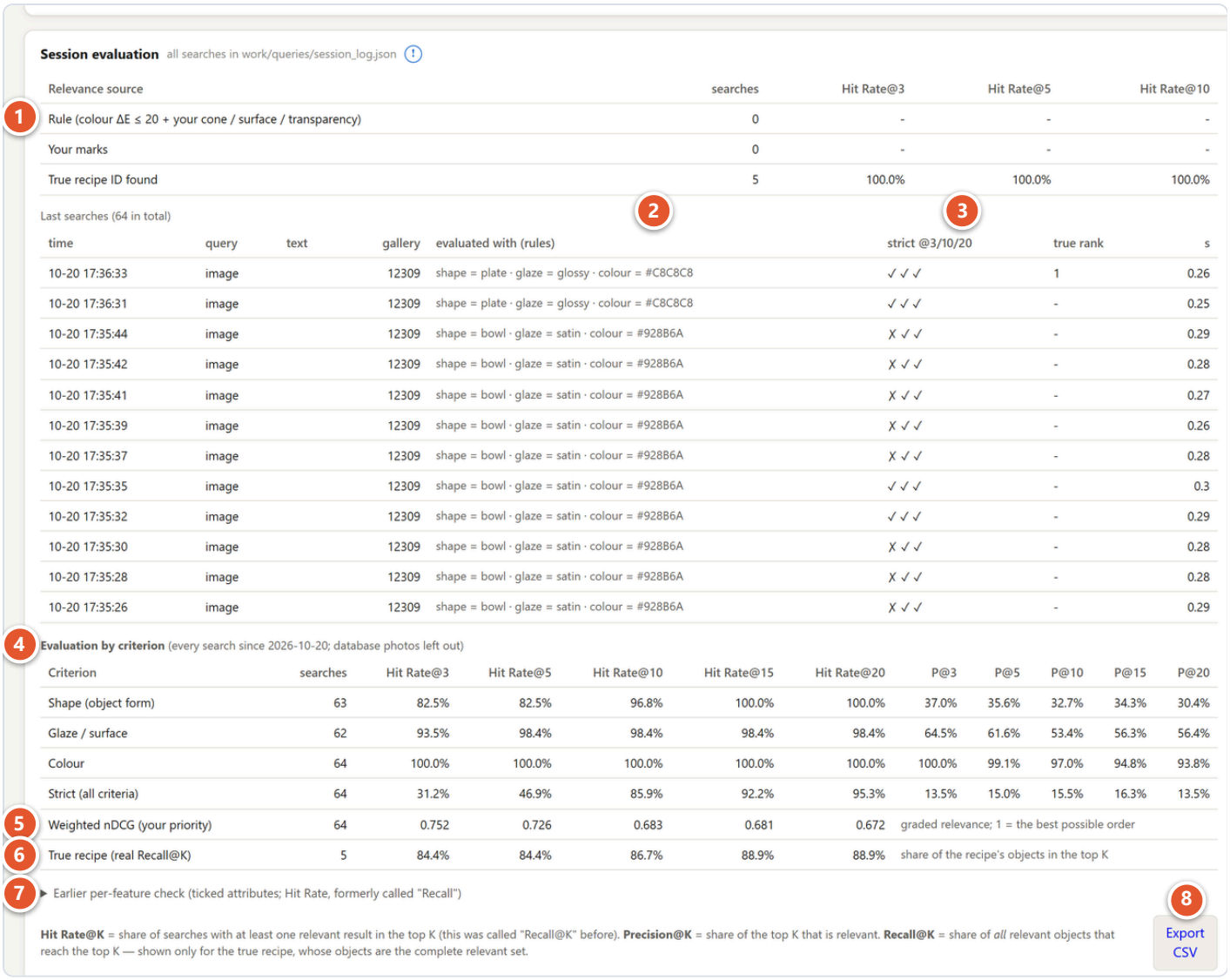
nDCG in three steps

1. Each result gets its weighted relevance (0 – 1), e.g. 0.80.
2. Lower ranks count less: rank r is divided by $\log_2(r + 1)$ (rank 1 ÷ 1, rank 3 ÷ 2, rank 7 ÷ 3).
3. The sum is divided by the best sum possible with the whole gallery in perfect order.

Renamed: before 2026-10-20 the session table called the hit rate "Recall@K". The numbers are the same, only the name is right now; the log itself is unchanged.

"✓ yes" still does not mean your object was found. The criteria say whether the results *look alike*. To check whether **the** right recipe is found, type its number in **True Glazy recipe ID**.

17 · Session evaluation



Session evaluation all searches in work/queries/session_log.json

	Relevance source	searches	Hit Rate@3	Hit Rate@5	Hit Rate@10
1	Rule (colour ΔE ≤ 20 + your cone / surface / transparency)	0	-	-	-
	Your marks	0	-	-	-
	True recipe ID found	5	100.0%	100.0%	100.0%

Last searches (64 in total) — 2 — 3

time	query	text	gallery	evaluated with (rules)	strict @3/10/20	true rank	s
10-20 17:36:33	image		12309	shape = plate · glaze = glossy · colour = #C8C8C8	✓ ✓ ✓	1	0.26
10-20 17:36:31	image		12309	shape = plate · glaze = glossy · colour = #C8C8C8	✓ ✓ ✓	-	0.25
10-20 17:35:44	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.29
10-20 17:35:42	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.28
10-20 17:35:41	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.27
10-20 17:35:39	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.26
10-20 17:35:37	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.28
10-20 17:35:35	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✓ ✓ ✓	-	0.3
10-20 17:35:32	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✓ ✓ ✓	-	0.29
10-20 17:35:30	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.28
10-20 17:35:28	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.28
10-20 17:35:26	image		12309	shape = bowl · glaze = satin · colour = #928B6A	✗ ✓ ✓	-	0.29

4 **Evaluation by criterion** (every search since 2026-10-20; database photos left out)

Criterion	searches	Hit Rate@3	Hit Rate@5	Hit Rate@10	Hit Rate@15	Hit Rate@20	P@3	P@5	P@10	P@15	P@20
Shape (object form)	63	82.5%	82.5%	96.8%	100.0%	100.0%	37.0%	35.6%	32.7%	34.3%	30.4%
Glaze / surface	62	93.5%	98.4%	98.4%	98.4%	98.4%	64.5%	61.6%	53.4%	56.3%	56.4%
Colour	64	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	99.1%	97.0%	94.8%	93.8%
Strict (all criteria)	64	31.2%	46.9%	85.9%	92.2%	95.3%	13.5%	15.0%	15.5%	16.3%	13.5%
5 Weighted nDCG (your priority)	64	0.752	0.726	0.683	0.681	0.672	graded relevance; 1 = the best possible order				
6 True recipe (real Recall@K)	5	84.4%	84.4%	86.7%	88.9%	88.9%	share of the recipe's objects in the top K				

7 ▸ Earlier per-feature check (ticked attributes; Hit Rate, formerly called "Recall")

Hit Rate@K = share of searches with at least one relevant result in the top K (this was called "Recall@K" before). **Precision@K** = share of the top K that is relevant. **Recall@K** = share of *all* relevant objects that reach the top K — shown only for the true recipe, whose objects are the complete relevant set.

8 Export CSV

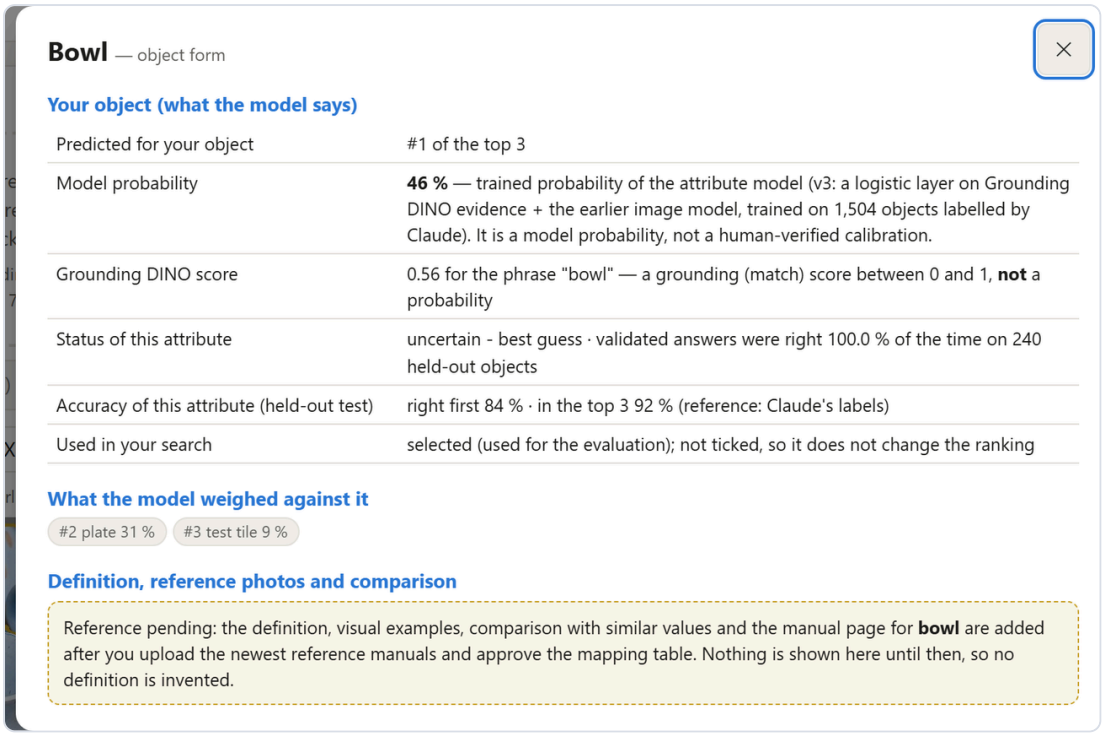
From the test copy used for this guide (its own log); your real log was not touched.

1. **Relevance sources** as before — Rule, Your marks, True recipe ID found — now as Hit Rate@3 / 5 / 10.
2. **Evaluated with (rules)**: the values each search was checked against.
3. **strict @3/10/20**: ✓ when a result matching all criteria was in the top 3 / 10 / 20.
4. **Evaluation by criterion**: Hit Rate@K and Precision@K over every search since 2026-10-20 (database photos left out).
5. **Weighted nDCG**, averaged over the searches.
6. **True recipe (real Recall@K)**, averaged over the searches with a True Glazy recipe ID.
7. **Earlier per-feature check** (folded): the old table, with Hit Rate as its name.
8. **Export CSV**: every search; the new columns are listed below.

> New CSV columns: `eval_mode`, `eval_priority`, `shape / glaze / colour / strict _Hit@3, 10, 20` and `_P@3, 10, 20`, `eval_nDCG@10`, `true_Recall@10`, `eval_rules`. Searches made before version 4 have no evaluation and are left out of "Evaluation by criterion"; they stay in the log and in the other tables. Everything is stored in `multimodal_experiments\work\queries\session_log.json`.

18 · The (!) and (i) explanations

(i) after every attribute value in card 1b opens what the model says about **exactly that value** for your object:



(i) after "bowl" (object form) for object #2 of the shelf photo.

Row	What it means
Predicted for your object	the place of this value in the top 3 (or "not in the top 3")
Model probability	the trained probability of the attribute model v3 (a layer on Grounding DINO evidence + the earlier image model, trained on 1,504 objects labelled by Claude) — a model probability, **not** a human-verified calibration
Grounding DINO score	how well the phrase matched the object, 0 – 1 — **not** a probability
Status / accuracy	validated, answered or uncertain; how often this attribute was right on 240 held-out objects (first / in the top 3)
Used in your search	ticked = changes the ranking; chosen but not ticked = used only by the evaluation
Definition, reference photos	"Reference pending" until your newest manuals are uploaded and the mapping table is approved

(!) buttons: the rules on the page itself

How object selection works

The photo can hold several ceramic objects. Each one has an outline from the segmentation (Grounding DINO finds it, SAM 2.1 traces it).

You do	What happens
Hover an object	it lights up (the rest dims) and a label shows its number, how sure CLIP is that it is ceramic, and its share of the photo
Click an object	it becomes the **active object** (thick yellow outline) and the only selected one
Ctrl + click	adds the object to the selection (it becomes active), or removes it; the other selected objects keep a white outline
Object list (details)	the same choices as a list; the yellow frame is the active object

Several selected objects - what is used where

Part of the page	Uses
Objects search (default)	only the **active** object: its look, shape, colour regions and attributes are compared with the 12,309 database objects
Card 1b (your object)	the active object
Recipes (previous model) search	**all selected** objects together (their image features are averaged by area)

> To compare two objects of the same photo, click the first, search, then click the second and search again.

Pick colour (HEX)

Switch the photo to **Pick colour** and click: the colour is averaged over a 7 × 7 pixel square, using **only the pixels inside the object under the cursor** (background is left out). You see the HEX value, can edit or copy it, and **Add as search colour** puts it in card 4 - it is used when card 4 is set to *typed / picked*. Clicking outside every object still gives a colour, with a warning.

Card 1: selection, several objects, Pick colour.

Priority of the photo part

The photo part of the score compares four things, grouped into three topics you can prioritise:

Topic	What is compared
Shape	outline geometry and the silhouette of the cut-out
Glaze / surface	glaze appearance (image model) + surface / effects (attribute probabilities)
Colour	the colour regions of the object (ΔE2000), only when "match colour" is ticked

The three always add up to 1. Raising one shrinks the other two **in proportion**; the mode buttons set validated presets:

shape 0.60 | glaze / surface 0.23 | colour 0.17

Now: shape 0.60 · glaze / surface 0.23 · colour 0.17 (custom)

Preset	Shape	Glaze / surface	Colour
Balanced	0.20	0.45	0.35
Shape priority	0.65	0.20	0.15
Glaze / surface priority	0.05	0.85	0.10
Colour priority	0.05	0.15	0.80

How it enters the final score

Each component's similarity is standardised over all searched objects (how far above average an object is) and multiplied by its weight. The photo part has 0.40 in total; ticked attributes in card 1b take 40 % of it; text adds 0.35 and a formula 0.25. All weights are divided by their total. Each result card shows the **contribution** of every component - their sum is the final score.

> Similarities are not probabilities. The sliders change the ranking only; the evaluation table never changes the order.

Search bar: the sliders, the presets and how they enter the score.

How each search is evaluated

Every search is checked on three criteria, using your object's values (the #1 candidate, or the one you clicked in card 1b). The labels of the database objects come from the best available source: your review > Claude's check > Grounding DINO > the earlier model.

Criterion	A result matches when
Shape	it has the same object form (e.g. cup)
Glaze / surface	its surface finish is in the same group (glossy / satin / matte); if you picked glaze effects, it also shares at least one of them
Colour	your most important colour is within ΔE2000 20 of one of its colour regions (or the same colour family, if ticked)
Strict	every criterion above matches (e.g. cup + glossy + brown). Cup + matte + brown is *not* fully relevant.
Weighted	your priority × matches, e.g. 0.20 × shape + 0.45 × glaze + 0.35 × colour (0 to 1)

? = the object has no label for that criterion: it is never counted as a match, and is shown apart.

> Evaluation never changes the ranking - the results are ordered by similarity; the evaluation only says how many of them are relevant.

Results: the criteria, strict, weighted and ?.

Hit Rate, Precision, Recall and nDCG

Metric	Question	Example (top 10 has 4 matching results)
Hit@K (one search)	is there at least one relevant result in the top K?	yes
Hit Rate@K (session)	in what share of my searches was Hit@K "yes"?	-
Precision@K	what share of the top K is relevant?	4 / 10 = 40 %
Recall@K	what share of *all* relevant objects reached the top K?	if 3,000 objects are cups: at most 10 / 3,000 = 0.3 % - not informative
nDCG@K	are the most relevant results (weighted by your priority) near the top? 1 = the best possible order	-

> Before 2026-10-20 the session table called the hit rate "Recall@K". It is renamed; the log itself is unchanged. True Recall@K is reported for the **True Glazy recipe ID**: that recipe's objects (usually 1 - 3) are the complete relevant set.

Session: Hit Rate, Precision, Recall, nDCG.

19 · Cheat sheet

Goal	Card 1 photo	Card 2 text	Mode / sliders	Tick in 1b	Extra
Find this exact glaze / recipe	the original photo	empty	Balanced	colour	True Glazy recipe ID if you know it
Same glaze on any object	photo	empty	Glaze / surface	surface finish, glaze effects (choose from top 3)	
Anything in these colours	photo, or none	empty	Colour	colour	mark important regions; or pick / type colours (card 4)
Same kind of object	photo	empty	Shape	object form	
Same kind of object, similar glaze too	photo	empty	Balanced, shape slider up (e.g. 0.60)	colour	the tag says "custom"
One object of a group photo	click it (yellow)	empty	any		then click the next one and search again
Colour + object + effect combined	photo	empty	Balanced	colour + object form + glaze effects	check each choice with (i)
Recipes described by facts	none	subtype + cone + surface + notes	—	—	Recipes (previous model) + hard filters
Similar chemistry	none	empty	—	—	card 3 oxides + ingredients

Common mistakes

Mistake	What you see	Do instead
a single word in card 2 ("glaze")	results all "cone ? · surface ?", ⚠ too vague	leave card 2 empty
Ctrl + clicking two objects and expecting both to be searched	the results match only one	the Objects search uses the active (yellow) object; search each in turn
typed or picked colours left on "typed / picked"	results ignore your photo's colours	switch card 4 back to "from the photo"
a tiny screenshot or a framed crop	the cut-out includes the frame; weak glaze-look match	use the original photo, cropped to the object
a wrong attribute ticked	the right object drops (rank 77 for form = bowl)	tick only what you are sure of; check the top 3 with (i)
reading colour Precision 100 % as success	colour ✓ almost everywhere	for neutral colours most objects match — look at shape, glaze and strict
reading "✓ yes" as "found"	criteria match, the recipe is far down	use True Glazy recipe ID
hard filters in Objects search	nothing changes	they work in Recipes (previous model) only

Start: `cd /d "D:\NUS study\CG2\code_project"` then `RUN_GLAZE_SEARCH.bat` → http://127.0.0.1:8770 (keep the window open; close it before any other GPU job).